Unit 1 Quiz: Displaying One Variable

Name _____

Show your work.

1.1 - Frequency Tables, Dot Plots, and Histograms

1. **I make and read a frequency table and a dot plot.**

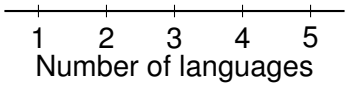
   16 students said how many languages they speak.

   2, 1, 3, 2, 4, 2, 1, 3, 2, 3, 2, 1, 5, 2, 3, 2

   (a) Make a frequency table.

Languages	Tally	Frequency
1		
2		
3		
4		
5		

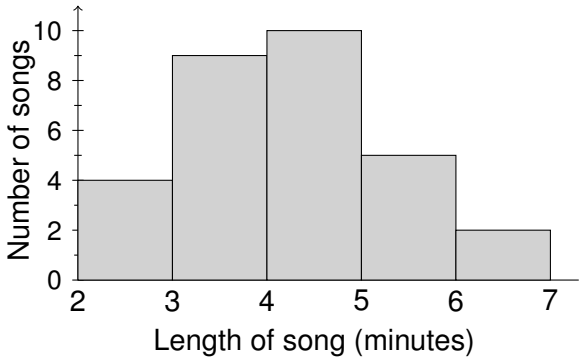
   (b) Make a dot plot.



   (c) How many students speak 3 or more languages?

2. **I read a histogram.**

   This histogram shows the lengths of 30 songs on a playlist.



   (a) How many songs are from 3 to 4 minutes long?

   (b) How many songs are 5 minutes long or longer?

   (c) One song is exactly 4 minutes long. Which bar is it in?

   (d) Can you tell from the graph how long the longest song is? Explain.

1.2 - Mean, Median, and Shape

3. **I find the mean and the median of a data set.**

A basketball player scored these points in 6 games.

14, 9, 21, 12, 9, 19

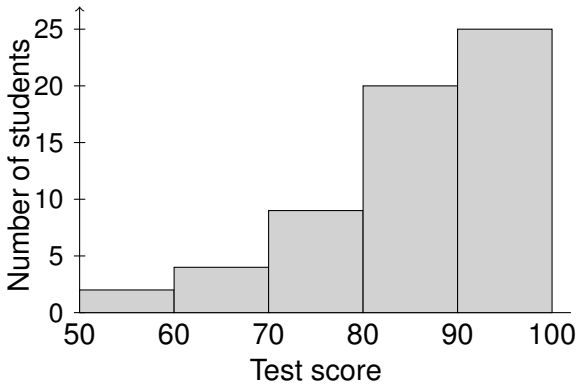
(a) Find Σx.

(b) Find the mean, $\bar{x}$.

(c) Find the median.

4. **I tell the shape of a graph, and I choose the best measure of center.**

This histogram shows the scores of 60 students on a test.



(a) Circle the shape of the data.

skewed left symmetric skewed right

(b) Which is greater, the mean or the median? Explain.

(c) Which statement is true? Circle the letter.

A The mean best describes a typical score, because the data are skewed left.
B The median best describes a typical score, because the data are skewed left.
C The mean best describes a typical score, because the data are symmetric.
D The median best describes a typical score, because the data are symmetric.

1.3 - Quartiles, IQR, and Box Plots

5. **I find the five-number summary and the IQR.**

11 students counted how many push-ups they did in one minute.

18, 25, 12, 30, 22, 15, 27, 20, 35, 16, 24

(a) Find the five-number summary.

Min _____ Q1 _____ Median _____ Q3 _____ Max _____

(b) Find the IQR.

6. **I make a box plot.**

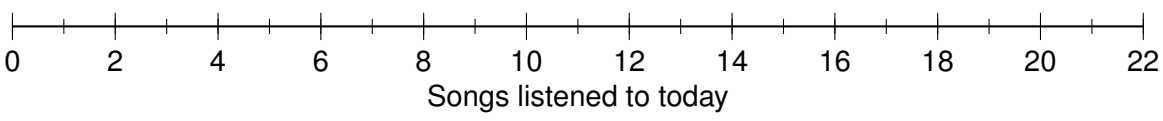
12 students counted how many songs they listened to today.

8, 3, 15, 10, 6, 12, 20, 9, 5, 14, 11, 7

(a) Find the five-number summary.

Min _____ Q1 _____ Median _____ Q3 _____ Max _____

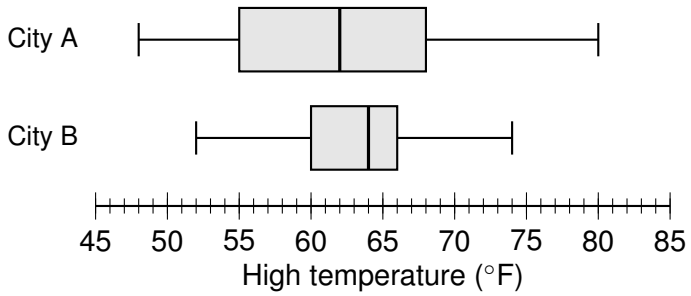
(b) Make a box plot.



1.4 - Reading Box Plots and Outliers

7. **I read and compare box plots.**

These box plots show the high temperature each day in April in two cities.



(a) Which city has the higher median?

(b) Find the IQR of each city.

(c) Which city has more consistent temperatures? Explain.

(d) About what percent of the days in City A had a high temperature above 55°F?

8. **I use the 1.5 × IQR rule to find outliers.**

9 customers recorded how many minutes they waited in line at a store.

13, 3, 18, 10, 34, 15, 12, 20, 16

(a) Find Q1, Q3, and the IQR.

(b) Find the lower fence and the upper fence.

(c) Which values are outliers? Explain.

(d) On a box plot of this data, where does the right whisker end? Why?